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RECESSED DOWNLIGHT FIXTURE

Abstract

A recessed downlight fixture and method for installation is provided. The recessed downlight fixture can be installed in a retrofit application, after a ceiling is installed, through a pre-existing opening in that ceiling. Alternatively, a recessed downlight fixture according to a second embodiment can be installed before a ceiling is present. In either embodiment, the light source coupled to, for example, a heat sink, can be universally moved in three degrees of movement, rotationally, along a tilt axis, or further within the ceiling to increase or decrease the recess. All such universal adjustments can take place through and below the ceiling opening during or after installation.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/541,541, filed Dec. 15, 2023; which is a continuation of U.S. patent application Ser. No. 17/734,393, filed May 2, 2022, now U.S. Pat. No. 11,846,401; which is a continuation of U.S. patent application Ser. No. 17/086,686, filed Nov. 2, 2020, now U.S. Pat. No. 11,302,101 issued May 3, 2022; which is a continuation of U.S. patent application Ser. No. 16/532,306, filed Aug. 5, 2019, now U.S. Pat. No. 10,823,349 issued Nov. 3, 2020; which is a continuation of U.S. patent application Ser. No. 15/088,922, filed Apr. 1, 2016, now U.S. Pat. No. 10,371,329, issued Aug. 6, 2019; all of which are incorporated by reference herein in their entirety. U.S. patent application Ser. No. 15/088,922 is related to co-pending applications filed concurrently therewith under U.S. patent application Ser. No. 15/088,813, titled “Recessed Downlight Fixture and Method for Installing and Universally Adjusting the Fixture in a Retrofit Application”, now U.S. Pat. No. 9,822,957, issued Nov. 21, 2017, and U.S. Patent Application Ser. No. 15,088/864, titled “Recessed Downlight Fixture and Method for Installing and Universally Adjusting the Fixture in a New Construction Application”, now U.S. Pat. No. 9,857,038, issued Jan. 2, 2018.

BACKGROUND OF THE INVENTION

[0002] 1. Field of the Invention

[0003] This invention relates to the field of interior lighting and, more particularly, to recessed downlight fixtures that can accommodate three degrees of adjustment below and through an opening in either a retrofit or new construction application.

2. Description of the Relevant Art

[0004] Lighting within a structure, such as a residential or commercial structure, is generally referred to as interior lighting. Contrary to interior lighting, lighting can also be employed exterior of the structure. Exterior lighting in many instances utilize a different lighting fixture and light source than an interior lighting fixture and source, and the exterior lighting device can be mounted altogether different from an interior lighting fixture and light source.

[0005] Similar to the differences between interior lighting and exterior lighting, interior lighting has many different types of lighting devices. The subject matter hereof is focused on interior lighting and, more particularly on a specific type of interior lighting, oftentimes referred to as recessed interior lighting.

[0006] Recessed interior lighting generally involves a recessed lighting fixture as well as a light source, where the fixture is typically installed in hollow ceiling spaces such that the fixture is hidden above an interior ceiling with only an opening and surrounding trim visible from below the ceiling. Recessed lighting fixtures are widely used for residential and commercial applications, and

generally comprise fixtures that fall within two broad categories: downlights or wall wash lights. In some instances, recessed lighting fixtures can fall within both categories of downlights and wall wash lights and are often grouped together and termed as downlights.

[0007] Downlights are designed so that the downlight fixtures direct light straight down from the ceiling, whereas wall wash lights and their associated fixtures cast an oblique illumination from the ceiling onto a nearby wall surface. The description hereof pertains to interior recessed lighting fixtures and, more specifically, interior recessed downlight fixtures.

[0008] Interior recessed downlight fixtures typically include a housing of some form of metal at least partially surrounding a light source and supported above an opening in the interior ceiling. The housing can be coupled to a source of electrical power with electrical conduit or the like pursuant to applicable building codes and regulations. The bottom of the recessed enclosure has an aperture which is aligned with an opening cut in the interior ceiling. The opening is normally finished with an ornamental trim which may also serve to support various accessories such as lens, light diffusers, condensers, baffles, filters and the like.

[0009] A common method of supporting recess light fixtures is by hanging those fixtures from existing ceiling joists, such as wooden beams. For example, a pair of parallel hanger bars can be nailed to the joist and the recessed housing can be hung between the two hanger bars by hanger brackets fastened to the recessed housing. The hanger brackets can be repositioned along the vertical line on the recessed fixtures so that the fixtures can be raised or lowered relative to the hanger bars, for adjusting the height of the recessed housing relative to the ceiling.

[0010] The recessed light fixtures are generally available in a variety of shapes, some of them being cylindrical, and others being square or rectangular boxes. The recessed light fixtures, while moveable on the parallel hanger bars between the ceiling joists, and also moveable in the vertical line for adjustment relative to the ceiling, the conventional recessed downlight fixture typically does not allow adjustment of the light source relative to the fixture. Instead, conventional recessed downlight fixtures only allow the light source to illuminate straight down along the vertical line once the fixture is secured to the hanger brackets.

[0011] The limited adjustment of a downlight fixture is increasingly problematic as the ceiling space or ceiling plenum becomes crowded. For example, the ceiling spaces can be occupied with ventilation ducts, fire sprinkler systems, conduits of various kinds for data cables, audio wiring, surveillance systems and layers of insulation. It sometimes occurs that the recessed interior downlight fixture cannot be installed in an optimal location because of such impediments in the ceiling space. Moreover, the junction box associated with the recessed downlight fixture oftentimes includes a transformer, ballast, or pulse width modulation (PWM) current drivers used, for example, as light emitting diode (LED) drivers that add to the crowded space above the recessed downlight fixture. Yet, the light source of a typical downlight fixture that has no adjustment capability must be directed to possibly an undesired location if the fixture is not optimally mounted on the hanger brackets and hanger bars.

[0012] Modern interior lighting has undergone a significant transformation in recent years. The incandescent light source, and in some instances the fluorescent light source, has been replaced by LEDs. More modern LED light sources generally include an array of LEDs thermally mounted to a heat sink. The LED drives obtain power from the AC mains. Recessed downlight fixtures that use LEDs are replacing existing incandescent or fluorescent recessed downlight fixtures at a rapid pace. One reason for replacing downlight fixtures with LED downlight fixtures is the energy savings, as well as the ability to control the color and luminance output from LEDs, which cannot easily be done in conventional incandescence or fluorescent light sources.

[0013] A need exists for retrofitting existing downlight fixtures with newer, more energy efficient LED recessed downlight fixtures. The desired LED recessed downlight fixtures can vary in structure and shape depending on whether they are installed in new construction or in an existing ceiling. In new construction, a ceiling does not exist, and the fixture can be mounted to the ceiling

joist. The ceiling is thereafter installed beneath the fixture with an opening through the ceiling to accommodate the light source and illumination therefrom. Conversely, in a retrofit application, the ceiling already exists. For example, the replacement LED downlight fixture must be inserted into the existing opening of the ceiling that previously accommodated the incandescent or fluorescent downlight fixture.

[0014] A need exists not only for installing a recessed downlight fixture in a retrofit application or new construction application, but also being able to make adjustments to after the install and even after the ceiling is in place. Regardless of whether the LED downlight fixture is utilized in a retrofit or a new construction application, such a fixture should beneficially be one that allows universal adjustment of the light source to overcome the increasingly crowded ceiling space and the constraints placed on the location of the recessed downlight fixture. This problem is even more acute when replacing incandescent or fluorescent recessed downlight fixtures with LED recessed downlight fixtures, since it is even more desirable to change the LED illumination pattern relative to the prior incandescent illumination pattern due to the difference in illumination between LEDs and incandescence or fluorescent lighting sources.

SUMMARY OF THE INVENTION

[0015] The problems outlined above are in large part solved by an improved recessed downlight fixture hereof. The recessed downlight fixture is improved in that it can accommodate an LED light source in either a retrofit application or a new construction application. In a retrofit application, the LED recessed downlight fixture is specifically adapted to replace the incandescent or fluorescent downlight fixture and use the space within the ceiling opening left by the removed incandescent or fluorescent downlight fixture. The retrofit application is applicable to the existing opening of the ceiling and without any modification to the ceiling or the previous installation brackets or hanger bars, for example. Conversely, in a new construction application, where the ceiling is not yet present, the improved LED recessed downlight fixture can be mounted to at least one ceiling joist and will allow universal adjustment to the light source similar in some ways to the retrofit application and its universal adjustment.

[0016] By allowing adjustment of the light source relative to the fixture and, more specifically, the fixture location within a ceiling and regardless of that location, the LED recessed downlight fixture can cast illumination from the LEDs in any adjustment pattern through the ceiling opening. In this fashion, the improved LED recessed lighting fixture hereof can not only retrofit to previously placed incandescent or fluorescent recessed lighting fixtures, but can compensate for changes in LED illumination from the incandescent or fluorescent illumination patterns as well as sub-optimal placement of the previous fixtures within the ceiling—either prior to the ceiling in a new construction application, or after the ceiling in a retrofit application.

[0017] According to a first embodiment, a recessed downlight fixture is provided in a retrofit application. The recessed downlight fixture, most preferably an LED recessed downlight fixture, comprises a foldable ring that can be inserted through an existing opening of the ceiling when placed in a folded configuration. Once the foldable ring is placed in a folded position through the ceiling opening, the foldable ring is expanded to an unfolded position to expose at least one ring protrusion extending radially inward from the foldable ring over a portion of the circular opening. A flange can then be secured by a coupling member to the ring protrusion, where the flange is then secured below and partially into the circular opening of the ceiling when the foldable ring is unfolded. Accordingly, the unfolded, foldable ring is placed above the ceiling and partially over the ceiling opening. The flange is secured below the ceiling and partially below the ceiling opening. The foldable ring and flange secured to the ring can accommodate a luminaire housing that is inserted through the ceiling opening and specifically through the opening formed by the ring and the flange.

[0018] The luminaire housing comprises a circular base on which the housing can move. The circular base comprises at least one radially inward extending recess dimensioned in the outer

circumference of the circular base to allow the ring protrusion to pass through the recess when the luminaire housing coupled to the circular base is passed through the opening from below the ceiling opening as well as the opening formed by the ring and attached flange. A pin is coupled to the circular base and is biased in a substantially circumferential direction against an upper surface of the ring protrusion to secure the circular base above the opening after the luminaire housing and circular base is extended to its further extent through the ceiling opening.

[0019] According to yet a further description of the first embodiment, the recessed downlight fixture in a retrofit application also comprises a first lever rotatably secured to the circular base and accessible solely within the opening and from below the opening within the ceiling. The first lever can be actuated by a user, when the user reaches into the ceiling opening from below the opening. When the first lever is rotated a first direction, the luminaire housing can rotate upon the circular base. When the first lever is rotated in a second direction opposite the first direction, the luminaire housing is prevented from rotating on the circular base.

[0020] According to yet a further description of the first embodiment, the recessed downlight fixture in a retrofit application includes at least one tilt arm having at least one tilt arm with first and second ends. A second lever is rotatably secured to a circular luminaire housing base and is accessible solely within the opening and from below the opening. When the second lever is moved in a first direction, the tilt arm allows the luminaire housing to rotate about the pin in a varying tilt position. The tilt position can vary to allow the light source to illuminate and at an angle offset from the perpendicular position relative to the ceiling. For example, the second lever, when actuated, can release the tilt arm and the light source coupled to the upper portion of the luminaire housing to be directed at an angle less than 90 degrees from the ceiling plane.

[0021] According to yet another feature of the first embodiment, the recessed downlight fixture in a retrofit application also comprises a post having a first end and a second end, where the first end is coupled to the luminaire housing and the second end extends into an aperture within a heat sink that accommodates a light source thermally bonded to the heat sink. Between the first end and the second end of the post is a groove into which an o-ring is placed. The o-ring frictionally engages between the post groove and the aperture within the heat sink to allow the heat sink and coupled light source to extend along a vertical line perpendicular to the ceiling when the heat sink and coupled light source is not tilted via the second lever. When tilted, the post allows the heat sink and coupled light source to recess further above the ceiling or less above the ceiling at an angle relative to the ceiling.

[0022] The combination of first and second levers as well as the slide able posts provides three degrees of adjustment of the upper portion of the luminaire housing relative to the opening within the ceiling. The upper portion preferably comprises a heat sink as well as a thermally bonded array of LEDs. By rotating via the first lever, tilting via the second lever, or recessing more or less via the posts, three degrees of adjustment can be performed by using the improved recessed downlight fixture within a retrofit application so as to accommodate a constrained space within the ceiling, the pre-existing ceiling opening and any angular configuration of the ceiling. By tilting the luminaire housing relative to the ceiling opening via the second lever, the downlight fixture can be used to maintain a straight downward illumination pattern even though the ceiling is pitched.

[0023] According to a second embodiment utilizing a recessed downlight fixture in a new construction application, a foldable ring need not be employed since the luminaire housing need not be inserted through an existing ceiling opening. Instead, the ceiling would not exist during installation, yet the ceiling would thereafter be installed below the fixture after the fixture is installed. The recessed downlight fixture according to the second embodiment can be one involving a plate having a planar surface configured to couple to at least one ceiling joist. The recessed downlight fixture according to the second embodiment also comprises a circular base coupled around an opening within the plate. A luminaire housing, and specifically a first portion of the luminaire housing, is rotatable within the circular base. A second portion of the luminaire housing

is tiltable relative to the first portion of the luminaire housing. A first lever is accessible through the opening within the plate for frictionally engaging the first portion of the luminaire housing to the circular base. A second lever is accessible also through the opening within the plate for frictionally engaging the second portion of the luminaire housing to the first portion of the luminaire housing. [0024] The recessed downlight fixture of the second embodiment also comprises a magnet coupled to a surface of a heat sink. A ferromagnetic member that can magnetically attract the magnet is coupled to the second portion of the luminaire housing, wherein the ferromagnetic member also comprises a first stop configured a first distance from the opening within the plate. A second stop is configured a second distance from the opening within the plate. The second distance being greater than the first distance. The heat sink can be moved by various means, such as for example, pressing upon the heat sink beneath the opening within the plate. Alternatively, the heat sink can be moved toward the opening within the plate. When the heat sink is moved toward the opening within the plate, the magnet is maintained against the first stop. When the heat sink is moved away from the opening within the plate, the magnet is maintained against the second stop.

[0025] Therefore, according to the second embodiment, the recessed downlight fixture is somewhat similar to the first embodiment recessed downlight fixture except that the second embodiment fixture is for new construction and the first embodiment downlight fixture is for the retrofit application. However, similar to the first embodiment retrofit downlight fixture, the new construction second embodiment downlight fixture includes a first lever and a second lever that frictionally engage to allow respective rotation of a luminaire housing relative to the opening and tilt adjustment of the luminaire housing relative to the ceiling opening. The first and second levers, like the first embodiment, frictionally engage to allow or disallow rotational and tilt adjustment. However, unlike the first embodiment of the retrofit application utilizing post and frictional engagement of posts within apertures of the heat sink, the second embodiment of a new construction application provides magnetic engagement instead of frictional engagement. The frictional engagement of posts with o-rings within apertures increases or decreases the amount of recess. The second embodiment performs the same increase or decrease in the amount of recess but using magnetic engagement.

[0026] According to yet a third embodiment, the recessed downlight fixture having a collar can be adjusted both rotationally and along a plane parallel to the ceiling. Specifically, according to the third embodiment, the collar of the fixture extends downward to form a circular flange. The flange that extends downward is one that extends perpendicular downward from a circular opening of the collar, where the flange extends within the circular opening of a ceiling to be formed about the flange after the fixture is secured to a ceiling joist. However, it is oftentimes desirable to align the circular flanges relative to one another within a ceiling. In many instances, an installer will ensure that all of the collars that are used to form the circular opening of the ceiling within a room are aligned using, for example, a laser aiming device.

[0027] To ensure the collars are all in a straight row, from beneath the plate, and specifically opposite the planar surface on which the circular base and luminaire housing are secured is a second plate. The second plate is movably secured to a planar surface of the plate and, more specifically, the collar used to form the ceiling opening, or ceiling aperture, extends through the plate as well as the second plate within the circumference of the circular base. The collar, however, is moveable with the second plate. When an installer aligns the collar so that all of the luminaire housings as well as the light sources are in line with one another, he or she moves the second plate which correspondingly moves the collar to the desired location on each plate. The installer can, for example, loosen one or more screws placed through the second plate and into at least one threaded washer that frictionally bears against the first planar surface of the plate on which the circular base and luminaire housing are mounted. In this fashion, the installer can adjust the collar, and specifically the ceiling opening around the collar so that all of the collars of each recessed downlight fixture in a new construction application can be aligned with one another in a single row,

for example. Moreover, the adjustment of the collar, like all adjustments in the first and second embodiments of the retrofit and new construction applications, are performed below the downlight fixture, for example, within a room beneath the joists or the ceiling. An installer either in a retrofit or new construction application can therefore easily adjust the downlight fixture, and specifically the luminaire housing on which the LED light source is thermally bonded in three degrees of movement, and all adjustments are performed beneath the downlight fixture, with most such adjustments being performed within the opening of the ceiling so that adjustments can be periodically made even after the ceiling is installed. Accordingly, three degrees of movement, or universal adjustment, in a retrofit or new construction application is purposefully made in LED light source installs to accommodate any LED illumination pattern within a downlight fixture.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] Other objects and advantages of the invention will become apparent upon reading the following detailed description and upon reference to the accompanying drawings.

[0029] FIG. 1 is a perspective view of a foldable ring inserted through a circular opening in a ceiling;

[0030] FIG. 2 is a cross sectional view of the foldable ring expanded above the circular opening upon which a circular flange is secured below and partially within the circular opening;

[0031] FIG. 3 is side view of a downlight fixture according to a retrofit embodiment, placed through the secured circular flange shown in cross section secured to an existing ceiling and toward a fixed recessed position above the ceiling;

[0032] FIG. 4 is a perspective view of the retrofit downlight fixture shown having a first lever moveable from inside the circular opening to allow rotation of a luminaire housing within the recessed downlight fixture in a plane parallel to the ceiling;

[0033] FIG. 5 is a partial cross sectional view of the luminaire housing within the retrofit downlight fixture moveable on pins extending substantially perpendicular to the ceiling when the luminaire housing is not tilted, and further having a second lever also moveable from inside the circular opening to allow tilt of the luminaire housing about at least two pins extending through the downlight fixture co-linear with one another, separated by the circular opening, and extending parallel to the ceiling;

[0034] FIG. 6 is a perspective view of a downlight fixture according to a new construction embodiment, secured between a pair of ceiling joist prior to a ceiling being applied to the joists;

[0035] FIG. 7 is a perspective view of the new construction downlight fixture showing a second lever similar to the second lever shown in FIG. 5 for the retrofit downlight fixture, wherein the second lever is moveable from inside the circular opening to allow tilt of the luminaire housing;

[0036] FIG. 8 is an exploded view of magnets coupled to opposite ends of a luminaire housing secured within the new construction downlight fixture to allow magnetic securement between upper and lower stops along pins that extend substantially perpendicular to the ceiling when the luminaire housing is not tilted;

[0037] FIG. 9 is a partial perspective view of the new construction downlight fixture having a first lever moveable from inside the circular opening to allow rotation of the luminaire housing within the recessed downlight fixture in a plane parallel to the ceiling;

[0038] FIG. 10 is a perspective view of the new construction downlight fixture having a collar configured to be placed into an opening of the ceiling but is rotatable about a central axis of the collar and moveable in a plane parallel to the ceiling to accommodate a fine-tune adjustment of the opening to other openings within the ceiling; and

[0039] FIG. 11 is a detailed view of cutouts in the plate also shown in FIG. 7, which holds the

luminaire housing and pins extending from the plate to prevent a threaded washer from moving when a second plate that holds the collar is moved in order to perform the fine tune adjustments of the collar opening to the room below.

[0040] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0041] Referring now in detail to the drawings, wherein like numerals indicate like elements throughout the several views. There is shown in FIGS. **1-5** a retrofit recessed downlight fixture according to a first embodiment. There is shown in FIGS. **6-9** a new construction recessed downlight fixture according to a second embodiment. There is shown in FIGS. **10** and **11** further details of the new construction recessed downlight fixture with moveable collar opening according to a third embodiment.

[0042] Beginning with the first embodiment, and referring to FIG. **1**, a foldable ring **10** is shown insertable through an opening in a ceiling **12**. The opening **14** results from the extraction of a previously placed recessed downlight fixture associated with, for example, an incandescent or fluorescent fixture in favor of a to-be installed LED recessed downlight fixture, for example. Alternatively, opening **14** could also be in a newly created opening in a ceiling where there may have not been any previously installed fixture and thus not necessarily used to replace an existing light fixture. In order to install the new LED recessed downlight fixture into opening **14**, retaining ring must be inserted into the opening **14** which pre-exists or is created within the ceiling **12** without disrupting ceiling **12**. Accordingly, ring **10** must be foldable upon itself.

[0043] Foldable ring **10** includes two axis **16a** and **16b** about which ring **10** folds onto itself. Ring **10** can either be of circular, square or rectangular outer dimension. Each axis can be formed by a pin **17a** and **17b**, respectively. The pins **17** and axis **16** are collinear with each other, as shown so that the ring can rotate about pins **17a** and **17b** to diminish the overall profile of the ring less than its diameter and, in fact, approximately equal to its radius so that ring **10**, when folded, can fit through the pre-existing opening **14** within ceiling **12**. Depending on the dimension of the ring, the opening **14** can be circular, square or rectangular. Included on ring **10** is at least one ring protrusion **18a**. As shown in FIG. **1**, there can be four ring protrusions **18a-d** that extend radially inward when the foldable ring is unfolded above the opening **14**, and above ceiling **12** in a retrofit application.

[0044] FIG. **2** is a cross-sectional view along plane 2-2 of FIG. **1**, and illustrates ring **10** unfolded upon the upper surface of ceiling **12**. Ring **10** is unfolded about the axes **16a** and **16b**, shown in FIG. **2**, and secured to a flange **20** placed below and partially into the circular opening **14**. FIG. **2** illustrates the unfolded, foldable ring **10** secured to flange **20** on opposite planar surfaces of ceiling **12**, all shown in cross-section. At least one coupling member **22** is placed through an aperture that extends radially inward from the flange **20** through an aperture within, for example, protrusion **18a**. The aperture within protrusion **18a** is shown as reference numeral **19a** in FIG. **1**. Also shown in FIG. **1** are respective apertures **19b-19d** corresponding to protrusions **18b-18d**. Coupling member **22** can be a screw, wherein the ring protrusion **18a-d** can comprise a threaded aperture **19a-19d** for receiving the screw **22**. For example, there can be four screws **22a-22d**, and the threaded screws **22a-22d** can be placed through apertures on flange **20** and into threaded apertures **19a-19d** on ring **10** to secure flange **20** onto ring **10**, both of which are held on opposing surfaces of ceiling **12**.

[0045] FIG. **2** illustrates a cross-sectional view along the plane 2-2 of FIG. **1**, and extends directly between protrusions **18a/d** and **18b/c** showing corresponding threaded screws **22a** and **22d** placed in respective apertures **19a** and **19d**. Similar to ring **10**, flange **20** is circular and of approximately the same interior dimension as ring **10**. Combination of ring **10** and secured flange **20** forms an

opening within the ceiling opening **14** to accommodate the luminaire housing and circular base. The luminaire housing coupled to the circular base is inserted below the ceiling **12** and through the opening formed by the interior dimensions of ring **10** and flange **20**.

[0046] Shown in FIG. **3** is a circular base **24** on which a luminaire housing **26** is moveably coupled. Luminaire housing **26** can comprise a first portion and a second portion of the luminaire housing, with a first portion **26a** sometimes interchangeably referred to as a circular luminaire housing base. It is understood, however, that luminaire housing **26** nonetheless moves relative to circular base, as does the first portion, or circular luminaire housing base **26a**. However, first portion **26a** rotates relative to circular base **24**, and second portion **26** can tilt relative to first portion **26a**, as well as circular base **24**. Moreover, second portion **26** can include a heat sink **28** that comprises a light source that moves perpendicular to the plane formed by first portion **26a** as well as the plane formed by circular base **24**, provided the second portion **26** is not tilted. FIG. **3** illustrates the first and second portions of the luminaire housing **26a** and **26**, respectively, as well as circular base **24**, inserted through the opening formed within ceiling **12** between the circular plate **10** and circular flange **20**, as shown by arrows **30**.

[0047] Within a circular heat sink **28** is at least one groove **32** that is dimensioned to allow the heat sink as well as all other portions of luminaire housing **26** and **26a** and the circular base to be inserted through the opening formed by plate **10** and flange **20**. When the luminaire housing **26** and its circular luminaire housing base **26a** is inserted entirely through the opening, circular base comprises at least one pin that is biased substantially circumferentially around circular base **24** and against an upper surface of the ring protrusion shown in dashed line **18**. Accordingly, the recess **32** accommodates the ring protrusion that extends into the recess **32** as the downlight fixture comprising the luminaire housing and circular base **34** are inserted through the opening. There can be more than one pin **34**, and the biasing member can comprise a spring around pin **34**. One end of the pin **34** has a sloped surface. That distal end and, specifically the slope **36** frictionally engages the ring protrusion as luminaire housing **26** is passed through the opening, causing the sloped end **36** to compress the spring on pin **34** and, once the protrusion passes through the angular distal end **36**, the spring on pin **34** will push the distal end outward so that the lower surface of the sloped distal end **36** will reside on the upper surface of the protrusion **18**. Once the sloped distal end extends outward in a circumferential direction about circular base **26**, the luminaire housing **26**, as well as the circular luminaire housing base **26a**, and circular base **24** are securely held by the ring **10** and flange **20** to ceiling **12**. Luminaire housing **26**, circular luminaire housing base **26a** and circular base **24** are held in a recessed position above ceiling **12**.

[0048] Turning now to FIG. **4**, an illustration is provided of the rotation of luminaire housing **26** upon a secured circular base **24**. In addition, FIG. **4** illustrates a first lever **38** secured to circular base **24** and used to frictionally engage the first portion of the luminaire housing **26a**, also known as the circular luminaire housing base.

[0049] The first portion of the luminaire housing, or circular luminaire housing base **26a** rotates within the circular base **24**. When lever **38** is moved upward from the position shown in dashed line to the position that is labeled, a block member shown in FIG. **5** within circular base **24** is configured to move upward between one end of first lever **28** and the circular luminaire housing base **26a**. Block member **40** is adapted to move between the first lever **38** and the circular luminaire housing base **26a** such that when it is compressed upward via the rotational movement of first lever **38**, it prevents rotation of the luminaire housing, and specifically the circular luminaire housing base **26a** within a plane parallel to the ceiling. When first lever **38** is moved downward, however, block member **40** also moves downward and no longer frictionally bears against the circular luminaire housing base **26a** so as to allow rotation of the luminaire housing in a plane parallel to the ceiling. The movement **42** shown in FIGS. **4** and **5** of first lever **38** to the position shown in dashed line is illustrative of member **40** moving away from and frictionally disengaging from the first portion of a luminaire housing, or circular luminaire housing base **26a**. Importantly, first lever

38 is accessed beneath the ceiling within the opening **14**, shown in FIG. **1** by a user standing within a room reaching up into the recessed downlight fixture to actuate the first lever **38** within the pre-existing opening of the ceiling.

[0050] FIGS. **4** and **5** also show the second portion of the luminaire housing, oftentimes referred to as simply the luminaire housing **26** having at least one tilt arm **44a-44d**. Each tilt arm **44** comprises opposed first and second ends. The first end is coupled to the circular luminaire housing base **26a** via a pin **46a-46d**. The second end of each tilt arm **44** is coupled above the luminaire housing base to the second portion of the luminaire housing, or luminaire housing **26**. The second end is coupled also by pins **48a-48d**, with only the pin **48b** showing in FIG. **4**.

[0051] A second lever **50** is shown in FIG. **5** coupled by a pin and cam member **52** to the circular luminaire housing base **26a**. When the second lever **50** is moved to the position shown in dashed line in FIG. **5** in the direction of arrow **54**, the cam member **52** extends against the tilt arm pin **46a**. Once the cam member **52** extends against the tilt arm pin **46a**, movement of the tilt arm pin **46a** is prevented and thereby preventing any movement to the tilt arm **44a** coupled to the tilt arm pin **46a**. Accordingly, as shown in FIG. **5**, movement of the second level **50** causes movement of a cam member **52**. Depending on the position of second lever **50**, the tilt arms can either rotate around their corresponding pins, or are prevented from rotating about their pins. If the tilt arms are allowed to move, then the second portion of the luminaire housing **26** can tilt on the first portion **26a**, as well as on circular base **24**. The tilt mechanism allows the second portion containing a heat sink **28** and a thermally bonded set of one or more LEDs **54** to illuminate at an angle offset from perpendicular at an angle less than 90 degrees from the plane formed by the circular luminaire housing base **26a**, circular base **24**, and ceiling **12**. The LED light source can comprise an array of LEDs of different colors, such as red, green, blue and white. The array of LEDs **54** can be encapsulated within an encapsulant material **56** and placed within a parabolic reflector **58** to form a PAR lamp, or PAR luminaire.

[0052] Shown in FIGS. **4** and **5** is second portion **26** comprising a heat sink **28** having at least one aperture **60a-60d**, with FIG. **5** showing two apertures **60b** and **60d**, in cross-section. Apertures **60** accommodate a post **62** and, if there are four apertures, four posts **62a-62d**. Each post has a first end and a post second end. The first end is coupled to the luminaire housing **26**, and specifically, the second portion of the luminaire housing **26**. Between the first end and the second end is at least one groove that extends around the circumference of each pin **62**. Within each groove is an o-ring and, if there are two grooves, two o-rings **64** and **66** for each post **62**. The aperture **60** within the heat sink allows the post **62** to slide up and down within the aperture, with the corresponding pair of o-rings **64** and **66** frictionally engaged within the aperture **60**, between the aperture inner walls and the recesses within post **62**. The o-rings are preferably made of a rubberized material with elasticity so as to apply a biasing force between the post and inner walls of each aperture, and to maintain that biasing force at whatever position the posts are within the corresponding aperture.

[0053] The luminaire housing **26**, and specifically the heat sink portion of the luminaire housing **26** containing the thermally bonded light source **54** moves up and down upon the posts **62** so as to increase the amount of recess above the ceiling, or decrease the amount of recess above the ceiling, depending upon the user desired position.

[0054] The combination of FIGS. **1-5** illustrates the first embodiment for a recessed downlight fixture, and specifically a fixture to be used in a retrofit application. The method for installing the fixture that includes ring **10**, flange **20**, circular base **24**, first and second portions of luminaire housing **26a** and **26**, respectively, first and second levers **38** and **50**, the associated block and cam members of those levers, and the posts **62** and corresponding apertures **60** within heat sink **28** as well as light source **54**, etc., all comprise the recessed downlight fixture used in a retrofit application. The methodology for installing the retrofit recessed downlight fixture therefore includes folding ring **10** onto itself, then inserting the folded ring **10** through an opening **14** within a ceiling, then unfolding the ring **10** above the ceiling **12**. Thereafter, a circular flange **20** is

attached below and partially within the opening **14** to the unfolded ring **10**. A circular base **24** on which a luminaire housing (first and second portions) is inserted into the opening formed by the unfolded ring and attached circular flange. The circular base in which the luminaire housing is moveable allows the luminaire housing to be moved in three degrees of motion using the first lever **38** to rotate the luminaire housing, the second lever **50** to tilt the luminaire housing, and the posts **62** to recess in a direction perpendicular to the ceiling. Importantly, all three degrees of motion can be actuated and frictionally stopped and held in a stationary position after movement through the opening from beneath the opening of the ceiling. Thus, once the recess downlight fixture is placed into the pre-existing opening **14** within ceiling **12** after, for example, removing a previously placed recessed downlight fixture, the present recessed downlight fixture that has been inserted can thereafter be adjusted in at least one of the three degrees of motion from beneath the opening and through the opening. No changes or modification whatsoever to the ceiling opening **14** are needed. [0055] FIGS. **6-9** illustrate a second embodiment in which a recessed downlight fixture is applied to a new construction application in which a ceiling is not yet present. Instead, all that is present during install is at least one ceiling joist **100**. Ceiling joist **100** is exposed from below, and either exists as a subfloor of the room above, or in the attic, along with ventilation ducts, fire sprinkler systems, conduits of various kinds for data cables, etc.

[0056] A recessed downlight fixture **102** for use in a new construction application can be coupled to a single ceiling joist **100a**, or coupled to a spaced apart pair of ceiling joists **100a** and **100b**, as shown in FIG. **6**. Fixture **102** includes a plate **104**, possibly having a junction box **106** and other elements of downlight fixture **102**, labeled **108**. Those other elements will be described in more detail with reference to FIGS. **7-9**. Similar to the first embodiment for the retrofit application, the second embodiment for a new construction application includes electrical components, such as wiring and AC mains interface. That interface is any electrical interface that can receive the AC mains power supply and convert that power supply to, for example, current drivers of an LED driver circuit. A portion of the AC mains interface can be contained within junction box **106**, with an appropriate cabling **110** between the junction box and the LED module. It is understood that similar interfaces and cabling are associated with the first embodiment, and need not be shown. [0057] Turning now to FIG. **7**, plate **104** is shown in more detail containing the various elements of the recessed downlight fixture, separate and apart from the junction box **106**, and cabling **110**. Placed on plate **104** are the other elements which, in combination with plate **104**, comprise the recessed downlight fixture **102**, also shown in FIG. **6**. Recessed downlight fixture **102** for new construction application therefore comprises plate **104** having opposed planar surfaces, with a first planar surface shown to receive a circular base. Circular base is secured to the first planar surface, which is the upper surface shown in FIG. **7** of plate **104**.

[0058] A luminaire housing **114**, and specifically a first portion of the luminaire housing **114a** on which a second portion **114b** extends above, rotates within the circular base **112**. While first portion of **114a** can rotate within circular base **112**, circular base **112** is rigidly fixed to plate **104**. The second portion **114b** is tiltable relative to the first portion **114a**. As such, the second portion **114b** of luminaire housing **114** can tilt relative to the first portion **114a** of luminaire housing **114**, and first portion **114a** can rotate relative to the plate. The overall luminaire housing **114** can therefore rotate and tilt relative to the plate **104**.

[0059] A first lever shown in FIG. **9** is accessible through the opening **120** within the plate **104** and, when actuated, frictionally engages the first portion **114a** of the luminaire housing **114** to the circular base to prevent further rotation. Absent movement of the first lever **122** along the arrow **124** in FIG. **9**, the first portion **114a** is free to rotate beyond 360 degrees, and up to approximately 365 degrees, within circular base **112**, and within the plane formed by circular base **112** substantially coplanar with plate **104**. A rotational extender **125** allows for an additional 5 degrees of rotation to when the extender **125** having a groove that allows the two protrusions to slide within that groove an additional 5 degrees of rotation.

[0060] FIG. 9 further illustrates in detail the first lever **122** and its upward or downward movement **124**. When moved upward, for example, one end of lever **122** bears against the circular base **112** about pivot pins **126**. When moved downward in the illustration of FIG. 9, first lever **122** will rotate downward thereby causing the end next to the pivot pins **126** to frictionally disengage from the circular base. Thus, while the pivot pin and lever **122** are fixed to the first portion **114a**, movement of the first lever **122** will cause frictional engagement or disengagement from the circular base **112** that is coupled to the stationary, non-moveable plate **104**. Thus, FIG. 9 illustrates the rotational movement, and fixing of that rotational movement at the desired rotational angle using a first lever **122**. The illustration of FIG. 9 provides a detail of that mechanism absent the overlying second portion that could, if shown, obstruct the rotational movement mechanism, and the fixing thereof by first lever **122**.

[0061] Referring to FIG. 7, a second lever **130** that is accessible through opening **120** within plate **104** can be moved to frictionally engage the second portion **114b** of luminaire housing **114** to the first portion **114a**. When second lever **130** is placed in a first position, tilt arms **132** are free to rotate about pins configured at one end of each tilt arm **132**. When the second lever **130** is placed in a second position, one end of second lever **130** bears against, either directly or indirectly, a pin **134** at one end of a tilt arm **132** to prevent movement of all of the tilt arms **132** relative to a pin rotatably coupled to first portion **114a**. Thus, the second lever **130** is moveable to secure at least one tilt arm attached to the second portion **114b** from tilting relative to the first portion **114a**. When second lever **130** does not bear against a pin **134** to therefore allow tilt, the second portion on which a heat sink **140** is coupled can tilt the light source from an angle that is as much as 45 degrees from a perpendicular angle. The perpendicular angle being an angle perpendicular to the planar surface of plate **104**.

[0062] The second portion **114b** of the luminaire housing **114** includes a bracket **144** that surrounds and secures heat sink **140**. Moreover, bracket **144** is coupled to the upper ends of tilt arms **132** via pins **134**, as shown. When the pins **134** are secured, and the tilt arms **132** cannot move, bracket **144** as well as heat sink **140** and the thermally bonded LED light source remains fixed in a tilted position relative to plate **104**. If second lever **130** does not bear against a pin **134**, then bracket **144**, heat sink **140** and light source bonded thereto are free to move in any tilt position offset from a perpendicular within a range of almost 45 degrees from perpendicular. FIG. 8 illustrates, in exploded view, a magnet **150** secured to at least one sidewall surface of heat sink **140**. Magnet **150** can be secured by brackets, screws, etc., as shown by items **152**. A ferromagnetic member **154** can be coupled to the second portion **114b** of the luminaire housing **114** and, specifically, the bracket **144** shown in FIG. 7.

[0063] Ferromagnetic member **154** can be made of any material that can magnetically attract magnet **150** to either a first stop **156** or a second stop **158**. First stop **156** is configured a first distance from the opening **120** in plate **104**, whereas the second stop **158** is configured a second distance from the opening **120**. The second distance is greater than the first distance. FIG. 7 illustrates a partial view of the ferromagnetic member **154**, coupled to bracket **144** of the second portion **114b**, whereas magnet **150** is coupled to a sidewall surface of heat sink **140**.

[0064] When a user pushes from within opening **120** the heat sink **140** away from the opening, magnet **150** comes to bear against second stop **158**. Because magnet **150** is magnetically attracted to the ferromagnetic material of ferromagnetic member **154**, the magnet maintains the heat sink **140**, as well as the light source bonded thereto, at the second distance further away from the opening than a first distance in which the heat sink would appear if magnet **150** were placed against the first stop.

[0065] While a user can move the light source away from or towards the opening **120**, magnetic **150** will maintain that light source in either a first distance or a second distance from the opening **120**. Accordingly, the amount of recess of light source can be moved and magnetically retained at the moved-to position. Contrary to the first and second degrees of movement and the frictional

retention using the first and second levers, the third degree of movement (or amount of recess) is magnetically maintained.

[0066] FIGS. **6-9** illustrate the recessed downlight fixture according to the second embodiment, whereby the fixture can be moved in three degrees of movement, with the amount of recess being magnetically maintained rather than frictionally maintained in the third degree of movement. Moreover, the second embodiment shown in FIGS. **6-9** provide the universal, or three degrees of movement, from beneath the plate **104** through an opening formed in the plate **104**. Still further, the second embodiment shown in FIGS. **6-9** are applicable to a new construction application in which the recessed downlight fixture is installed prior to an installation of the ceiling, and which allows universal, three degrees of adjustment to accommodate movement of the light source either rotationally, tiltably, and/or an amount of recess, to accommodate the subsequently placed ceiling, the angle of that ceiling, the type of ceiling, the space above the ceiling, the light source and the various types of illumination achieved by any type of light source.

[0067] FIGS. **7, 10** and **11** illustrate a third embodiment in which a collar opening can be adjusted in a planar x-y direction abutting against plate **104**. Moreover, the collar can be rotated within the plane of plate **104**. The purpose of the third embodiment is to allow an installer of the recessed downlight fixture to move the collar so that, for example, all of the collars within any given room can be placed at any desired location preferably in a straight row among a group of collars, and with less than an inch of variation.

[0068] FIG. **7** illustrates opening **120** formed by a collar **200**. Collar **200**, similar to all of the other components of a recessed downlight fixture preferably comprises a non-reflective material such as, for example, anodized aluminum that is of a substantially darkened color, such as black. Collar **200** includes a first portion that is co-planar with plate **104**, and a second portion that extends perpendicular to the first portion, and downward from the first portion and the plane formed by plate **104**. The second portion of collar **200** is circular, and extends downward at a perpendicular angle approximately one inch, or any distance comparable to the thickness of a ceiling. The ceiling would be installed around the downward extending portion of collar **200**. For example, if the ceiling is made of drywall material, or gypsum board, the distance at which the second portion extends downward can be approximately one inch. It is recognized that the downward extending portion can be greater than an inch or less than an inch depending upon the thickness of the ceiling, however.

[0069] As shown in FIG. **7**, the second planar surface of the plate **104** includes cutouts **202** in plate **104**. Cutouts **202** extend to an opening within plate **104**, also shown in FIG. **11**. Extending from the lower planar surface of a second plate that slidably moves within a plane below plate **104** is a screw, or any other form of coupling member **206**. On the threads of screw **206** is a threaded nut on a washer, both of unibody construction. The nut and washer are shown as reference numeral **208**. Washer **208** extends beyond the cutout **202**.

[0070] As shown in FIG. **10**, the opposite planar surface from that of the threaded washer **208** is shown having a second plate **212**. The second plate **212** includes at least one aperture through which screw **206** extends. The aperture is of a diameter approximately equal to the diameter of screw **206** so that when the head of screw **206** is loosened, screw **206** moves to loosen the frictional abutment between washer **208** and plate **104**. Once screw **206** is loosened to a sufficient degree, then second plate **212** can move within a plane shown by arrows **214** and **216**, substantially coplanar and abutting plate **104**. The degree of movement of second plate **212** depends upon the amount of cutout **202** beneath threaded washer **208**. Screw **206** can move to the edges of cutout **202** when second plate **212** is moved. Pin **220** extends from the second plate **212** to prevent rotational movement of threaded washer **208**, when the head of screw **206** is loosened from beneath plate **104**. Accordingly, at least one screw **206**, and preferably four screws can be loosened and, once loosened, second plate **212** can move in the x and y directions as shown by arrows **214** and **216** on the lower surface of plate **104**. Once the collar **200** that is coupled to the second plate **212** is

moved to the appropriate position between, for example, the ceiling joists and to where the opening of the ceiling will be configured, screws **206** are tightened, and the corresponding threaded washer **208** will compress down upon plate **104** to prevent any further movement of the collar within the x-y plane. In this fashion, the collar can be placed wherever needed within the confines of the cutout **202** so that when an installer installs the recessed downlight fixture, he or she can then align one or more collars of associated fixtures precisely within a room using, for example, a laser aiming tool, or otherwise. As shown in FIG. **10**, collar **200** can also be rotated along the arrow **224**. Once rotated to the appropriate position, screws can be inserted into the first portion of the collar **200** and against the second plate **212**, as shown by reference numerals **226**.

[0071] The recessed downlight fixture according to the third embodiment therefore includes a luminaire housing that is rotatable within the circular base and tiltable relative to the circular base, and also includes a second plate moveably secured on the second planar surface of plate **104** and coupled to a collar **200** dimensioned to form an aperture that extends through the plate and the second plate as well as the circular base **112**. The collar **200** is moveable within a plane, as well as rotatably moveable between the second plate **212** and the luminaire housing by at least one screw **226** configured to be placed into a threaded opening in the collar and frictionally engage against the second plate. At least one screw **206** that is placed through the second plate is placed at least into a threaded washer **208** that when the screw is tightened, the threaded washer **208** frictionally bears against the first planar surface of plate **104**. The collar is moved from below the plate while standing in a room, for example. The collar is moveable in two dimensions (e.g., x-y directions) as well as rotational within a plane parallel to the second planar surface.

[0072] It will be appreciated to those skilled in the art having the benefit of this disclosure that this invention is believed to provide an improved downlight fixture that can accommodate any recessed application, whether that application is in new construction prior to a ceiling being installed or in a retrofit after the ceiling is present. The downlight fixture can also accommodate any ceiling space, whether tilted or not, or any recessed or rotational amounts needed for that space. Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. It is intended that the following claims be interpreted to embrace all such modifications and changes. Accordingly, the specification and drawings are to be regarded in an illustrative, rather than a restrictive, sense.

Claims

1. A recessed downlight fixture comprising: a plate, wherein the plate comprises a circular cutout and a first planar surface; a circular base, seated in the circular cutout of the plate and coupled to the first planar surface of the plate; and a luminaire housing comprising a first portion and a second portion, wherein the first portion of the luminaire housing is fixed to the circular base such that the first portion of the luminaire housing can rotate in a plane defined by the circular base, wherein the second portion of the luminaire housing is affixed to the first portion of the luminaire housing such that the second portion of the luminaire housing can tilt relative to the first portion of the luminaire housing.
2. The recessed downlight fixture of claim 1, wherein the circular base is coupled to the first planar surface of the plate using screws and washers.
3. The recessed downlight fixture of claim 1, further comprising a first lever, secured to the circular base, wherein the first lever can be actuated to frictionally engage the first portion of the luminaire housing to the circular base, to prevent the first portion of the luminaire housing from rotating in the plane defined by the circular base.
4. The recessed downlight fixture of claim 3, wherein the first lever is accessible through an opening, wherein the opening is defined by the circular cutout.
5. The recessed downlight fixture of claim 3, wherein, the first lever can be actuated between an

engaged position and a disengaged position, wherein, in the engaged position, the first lever frictionally engages the first portion of the luminaire housing to the circular base to prevent the first portion of the luminaire housing from rotating in the plane defined by the circular base, and wherein, in the disengaged position, the first lever is disengaged from the first portion of the luminaire housing from the luminaire housing, allowing the first portion of the luminaire housing to rotate in the plane defined by the circular base.

6. The recessed downlight fixture of claim 1, further comprising a second lever, wherein the second lever can be actuated to frictionally engage the second portion of the luminaire housing to the first portion of the luminaire housing, to prevent the second portion of the luminaire housing from tilting relative to the first portion of the luminaire housing.

7. The recessed downlight fixture of claim 6, wherein the second lever is accessible through an opening, wherein the opening is defined by the circular cutout.

8. The recessed downlight fixture of claim 6, wherein the second portion of the luminaire housing is affixed to the first portion of the luminaire housing by a plurality of tilt arms, each respective tilt arm of the plurality of tilt arms being affixed to the first portion of the luminaire housing by a respective pivot pin, such that each respective tilt arm of the plurality of tilt arms arm can rotate about its respective pivot pin, enabling the second portion of the luminaire housing to tilt relative to the first portion of the luminaire housing.

9. The recessed downlight fixture of claim 8, wherein the second lever, when actuated to a first position, bears against the one or more pivot pins affixing the one or more tilt arms to the first portion of the luminaire housing, to prevent rotation of the one or more tilt arms, and when actuated to a second position, is disengaged from the one or more pivot pins, allowing rotation of the one or more tilt arms about the pivot pins.

10. The recessed downlight fixture of claim 1, wherein the second portion of the luminaire housing further comprises a ferromagnetic member and a heat sink, wherein a magnet is affixed to a sidewall surface of the heat sink, wherein the heat sink can be actuated between a first stop and a second stop, wherein the heat sink, when actuated to the first stop is magnetically engaged with a first end of the ferromagnetic member, and the heat sink when actuated to the second stop, is magnetically engaged with a second end of the ferromagnetic member.

11. The recessed downlight fixture of claim 10, wherein the first stop is a first distance from an opening defined by the circular cutout, wherein the second stop is a second distance from the opening, wherein the first distance is different than the second distance.

12. The recessed downlight fixture of claim 10, wherein the magnet is affixed to the sidewall surface of the heat sink using brackets and screws.

13. The recessed downlight fixture of claim 10, further comprising a light emitting diode (LED) light source, the LED light source being thermally bonded to the heat sink.

14. The recessed downlight fixture of claim 1, further comprising a rotational extender coupled to the first portion of the luminaire housing.
